

Algebra II with Trigonometry

Chapter 5.4

Olympiad Problems (Gemini)

Saved tutoring response with MathJax rendering. Original PDF file:
[4401270_alg-2trig-h-chp-5-4-note-docx.pdf](#)

Here are 6 challenging, Olympiad-style mental math problems inspired by the concepts of inverse trigonometric domains, right triangle applications, and nested functions found in the text.

Problems

1. Let $\theta_n = \tan^{-1} \left(\frac{1}{\sqrt{n}} \right)$. Evaluate the exact value of the product:

$$\cos(\theta_1) \cdot \cos(\theta_2) \cdot \cos(\theta_3) \cdots \cos(\theta_{99})$$

2. Evaluate the exact value of the expression below (where the angle 5 is measured in radians):

$$\sin^{-1}(\sin 5) + \cos^{-1}(\cos 5)$$

3. Evaluate the exact value of the following nested composition, expressing your final answer in radians:

$$\cos^{-1} \left(\sin \left(\tan^{-1} \left(\frac{1}{\sqrt{3}} \right) + \tan^{-1}(1) \right) \right)$$

4. Using the reciprocal properties of inverse trigonometric functions, evaluate the exact value of:

$$\sin \left(\sec^{-1} \left(\frac{5}{4} \right) + \csc^{-1} \left(\frac{5}{3} \right) \right)$$

5. Find the exact positive real number x that satisfies the equation:

$$\sin^{-1}(x) + \sin^{-1}(2x) = \frac{\pi}{2}$$

6. In a right triangle with a hypotenuse of length 10, the two acute angles α and β satisfy the equation:

$$\tan^{-1}(\sin \alpha) + \tan^{-1}(\cos \beta) = \tan^{-1}\left(\frac{8}{15}\right)$$

Find the exact length of the triangle's shortest leg.